

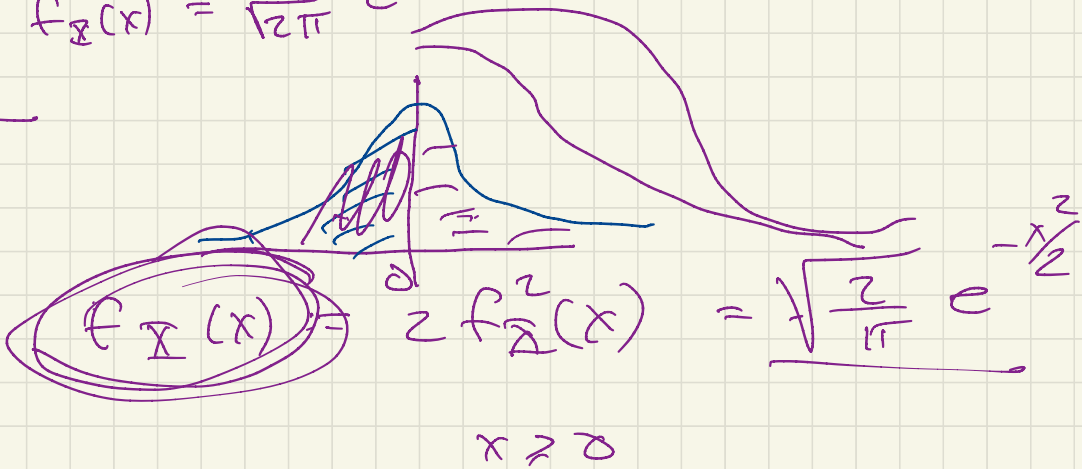

A-R - Normal (μ, σ^2)

S, $\bar{X} \sim N(0, 1)$ ent

$$Y := \mu + \sigma \bar{X} \Rightarrow Y \sim N(\mu, \sigma^2)$$

$$f_{\bar{X}}^Z(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Obs



Candidate a $Y \sim \text{Exp}(1)$, i.e.

$$g(x) = e^{-x}$$

$$h(x) = \frac{f(x)}{g(x)} \Rightarrow c = \sqrt{\frac{2e}{\pi}}$$

Algori

Algoritmo Normal A-B $X \sim N(\mu, \sigma^2)$

① Generar $Y \sim \text{Exp}(1)$.

② Generar $U \sim (0,1)$.

③ Si: $U \leq \frac{f(Y)}{c g(Y)} \Rightarrow Z = Y$

y vamos al paso ④. En otro caso
regresamos al paso ①.

④ $W = \begin{cases} Z & \text{con prob } Y/2 \\ -Z & \text{" " } Y/2 \end{cases}$

⑤ $X = \mu + \sigma W$

